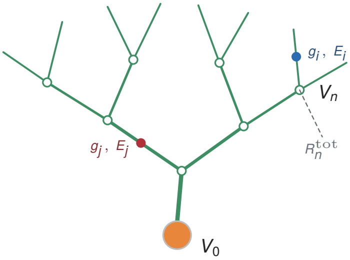
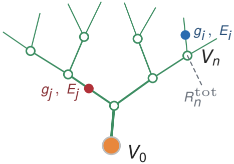
mode=forward: scale 1.0 vs 0.62

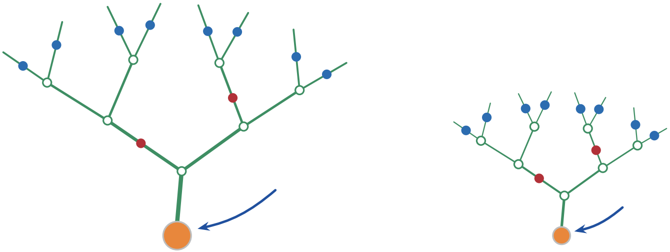


forward biophysics



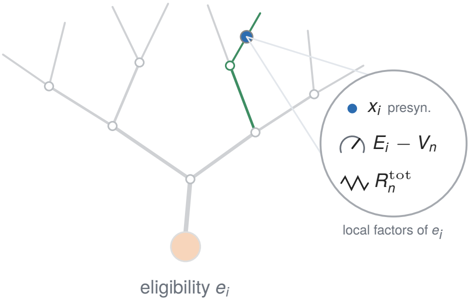
small

mode=plain: hook variant, synapse dots, no labels

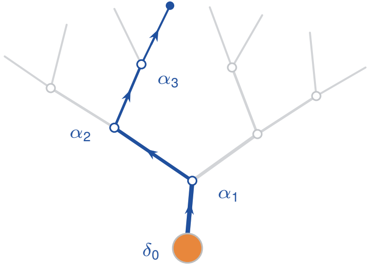


one output error arrives at the soma

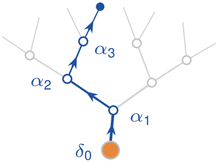
mode=eligibility: zoom bubble with three local factors



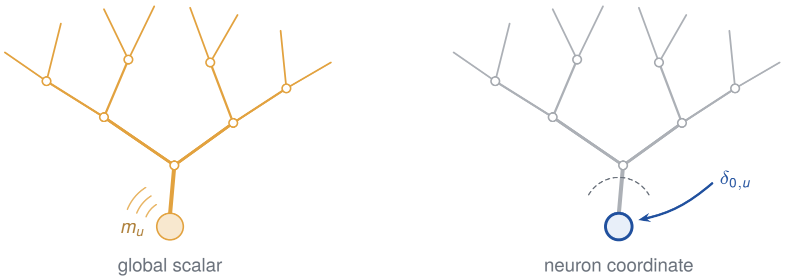
mode=transport: soma error along one root-to-leaf path



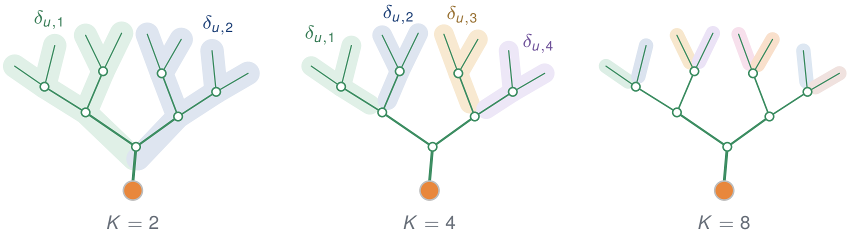
adjoint transport



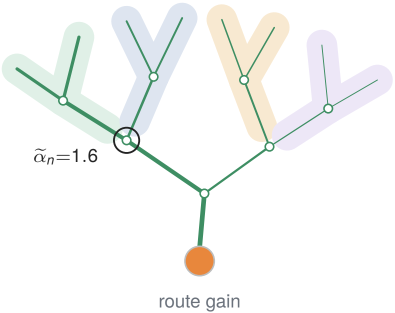
mode=scalar and mode=coordinate



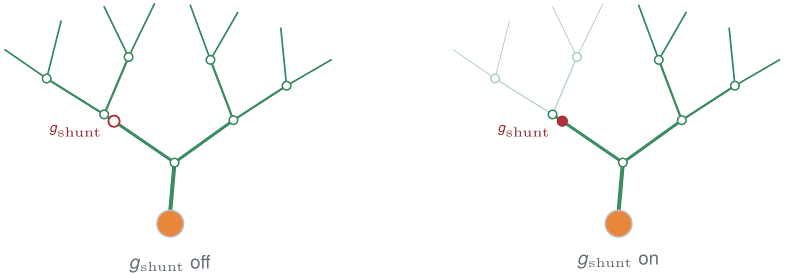
mode=address: K = 2, 4, 8



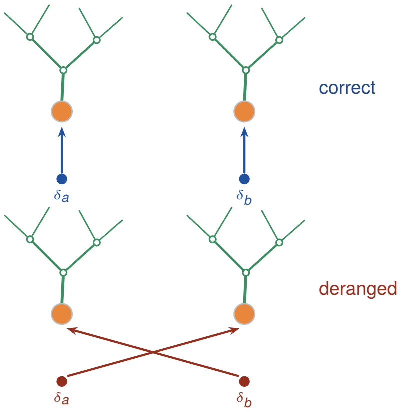
mode=gain: widths proportional to route gains



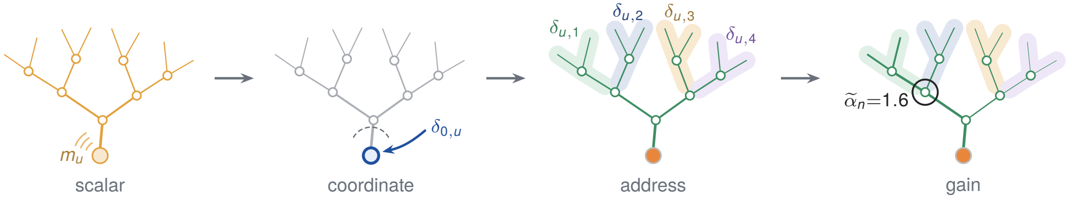
mode=shunt: shunted=false vs shunted=true



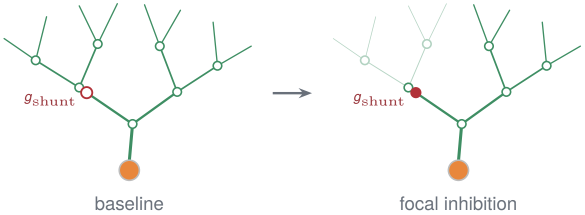
DerangedPair: correct vs crossed coordinates



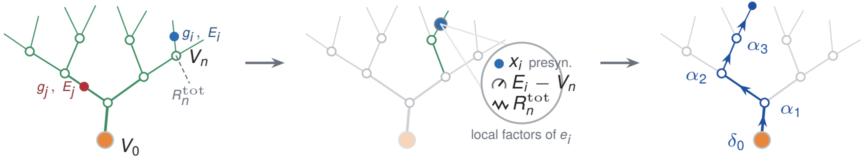
TreeStrip: scalar to coordinate to address to gain



TreeStrip: shunt off to shunt on



TreeStrip: forward to eligibility to transport

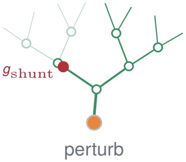


eqpanel, deftag, triptych

DEFINITION eligibility is the purely local factor.

DEFINITION

$$e_i = x_i\, R_n^{\mathrm{tot}}\, (E_i - V_n) \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial g_i} = e_i\, \delta_{0,u}\, \widetilde{\alpha}_{n(i)}$$



THEOREM

$$\widetilde{\alpha}' = \widetilde{\alpha}\, \frac{R'}{R}$$

The perturbed subtree's credit shrinks by the Sherman–Morrison factor; sibling routes are untouched. This column is the widest (0.36) and holds the experimental readout.

heroeq: factorization with adjacent term annotations

THEOREM

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{e_i}_{\text{local eligibility}} \; \underbrace{\delta_n}_{\text{transported error}} \qquad \delta_n = \delta_{0,u}\,\widetilde{\alpha}_n$$

Adjacent-term collision test: the two braces and their labels must stay clear of each other, baselines aligned.

heroeq: descent bound, three-term anatomy

BOUND

$$\mathbb{E}\,\mathcal{L}(\boldsymbol{w}^{+}) \le \mathcal{L}(\boldsymbol{w}) - \eta \underbrace{\boldsymbol{g}^{\top} M \boldsymbol{g}}_{\text{retained signal}} + \frac{L\eta^{2}}{2}\Big(\underbrace{\|M\boldsymbol{g}\|_2^2}_{\text{curvature bias}} + \underbrace{\operatorname{tr}(M \Sigma M^{\top})}_{\text{admitted noise}} \Big)$$

Eyebrow tag, Shunt left rule only (no box), colors per semantics: green retained, amber bias, BP-red noise.

heroeq: optimal attenuation, untagged variant

$$a_b^* = \min\left\{ 1,\; \underbrace{S_b}_{\text{branch signal}} \Big/ \underbrace{c\,(S_b + N_b)}_{\text{dose} \times \text{(signal + noise)}} \right\}$$

Untagged `\heroeq` at `\large`; slash division keeps both annotated terms at top display level.